Univariate Exploratory Data Analysis of NFL Dataset

David Mocianko

September 2025

1 Data Validation

Dataset used for this analysis contains information on past NFL regular season games.

Property	Value
Number of Rows	13500
Number of Columns	8
Missing Values	0 (0.00%)

Table 1: Dataset dimensions and quality checks for the NFL dataset.

The dataset has no missing values and is complete. The column names are also consistent and standardized.

2 Identify Variable Types

Each column in the NFL dataset was classified according to its variable type (numeric, categorical, or date).

Column Name	Type	Description
season	Integer	Year the season started
date	Date	Date of the game(YYYY-MM-DD)
team1	Categorical	Abbreviation for the first team (Home)
team2	Categorical	Abbreviation for the second team
result	Categorical	Game outcome for team1 (W,L,T)
score1	Numeric	Points scored by team1
score2	Numeric	Points scored by team2
home_away	Categorical	Indicates if team1 played Home or Away

Table 2: Classification of variables in NFL dataset

3 Descriptive Statistics

This section summarizes the numerical and categorical variables of descriptive statistics in the NFL data set. Numeric variables include (`score1`, `score2`) are analyzed using measures of central tendency, dispersion and distributional shape. Categorical variables (`result`, `home_away`, `team1`, `team2`) are analyzed using frequency counts and proportions.

3.1 Numeric Variables

Metric	**Score1**	**Score2**	**Pooled**
Number of Observations	13500	13500	27000
Mean	22.397	20.000	21.137
Median	22	20	20
Mode	20	17	17
Standard Deviation	10.491	10.153	10.399
Variance	110.052	103.075	108.146
Skewness	0.330	0.321	0.330
Kurtosis	0.063	-0.166	-0.031
Minimum	0	0	0
Maximum	72	62	72
1st Quartile (Q1)	15.0	13.0	14.0
2nd Quartile (Q2, Median)	22.0	20.0	20.0
3rd Quartile (Q3)	29.0	27.0	28.0
Interquartile Range (IQR)	14.0	14.0	14.0

Table 3: Descriptive statistics of numeric variables (`score1`, `score2`, and pooled).

3.2 Categorical Variables

Category	**Count**	**Proportion (%)**
Win (result = W)	7648	56.65 %
Loss (result = L)	5761	42.67 %
Tie (result = T)	91	0.67 %

Table 4: Counts and proportions for the variable `result`.

Category	**Count**	**Proportion (%)**
Home	13500	100 %
Away	0	0 %

Table 5: Counts and proportions for the variable `home_away`.

Team1	**Count**	**Proportion (%)**
LAC	458	3.39
NYG	458	3.39
PHI	458	3.39
MIN	458	3.39
DEN	457	3.39
TEN	457	3.39
BUF	457	3.39
PIT	457	3.39
IND	457	3.39
LAR	457	3.39
SF	457	3.39
GB	456	3.38
DET	456	3.38
ATL	456	3.38
ARI	456	3.38
MIA	456	3.38
CHI	456	3.38
LV	456	3.38
DAL	456	3.38
WAS	456	3.38
KC	456	3.38
NYJ	455	3.37
NE	455	3.37
NO	449	3.33
CIN	441	3.27
CLE	432	3.20
SEA	387	2.87
TB	386	2.86
JAX	242	1.79
CAR	242	1.79
BAL	234	1.73
HOU	186	1.38

Table 6: Counts and proportions for team abbreviations (`team1`)

Team2	Count	Proportion (%)
NYJ	458	3.39
NE	458	3.39
LV	457	3.39
GB	457	3.39
WAS	457	3.39
MIA	457	3.39
CHI	457	3.39
KC	457	3.39
DAL	457	3.39
ARI	457	3.39
DET	457	3.39
ATL	457	3.39
DEN	456	3.38
LAR	456	3.38
IND	456	3.38
TEN	456	3.38
PIT	456	3.38
SF	456	3.38
BUF	455	3.37
PHI	455	3.37
NYG	455	3.37
MIN	455	3.37
LAC	455	3.37
NO	450	3.33
CIN	443	3.28
CLE	433	3.21
TB	387	2.87
SEA	386	2.86
CAR	242	1.79
JAX	242	1.79
BAL	234	1.73
HOU	186	1.38

Table 7: Counts and proportions for team abbreviations (`team2`)

Team	**Count**	**Proportion (%)**
MIA	913	3.38
TEN	913	3.38
LAC	913	3.38
GB	913	3.38
ARI	913	3.38
ATL	913	3.38
DET	913	3.38
PIT	913	3.38
DAL	913	3.38
SF	913	3.38
WAS	913	3.38
LAR	913	3.38
LV	913	3.38
DEN	913	3.38
NYJ	913	3.38
MIN	913	3.38
NE	913	3.38
KC	913	3.38
PHI	913	3.38
IND	913	3.38
NYG	913	3.38
CHI	913	3.38
BUF	912	3.38
NO	899	3.33
CIN	884	3.27
CLE	865	3.20
SEA	773	2.86
TB	773	2.86
JAX	484	1.79
CAR	484	1.79
BAL	468	1.73
HOU	372	1.38

Table 8: Counts and proportions for team abbreviations (occurrences in `team1` and `team2` combined)

4 Visualizations

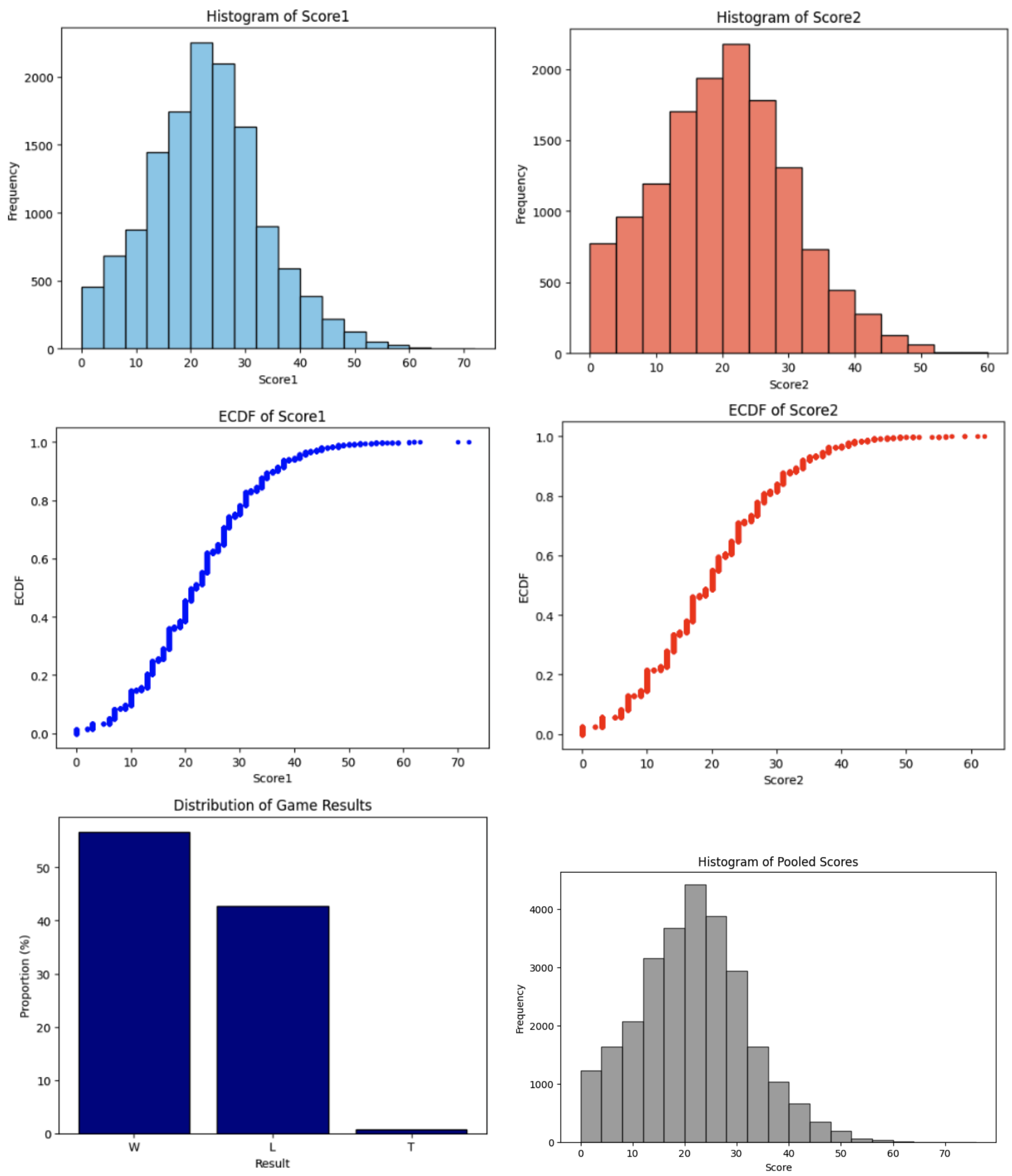

Figure 1: Descriptive visualizations of NFL game statistics: (a) Histogram of Score1, (b) Histogram of Score2, (c) ECDF of Score1, (d) ECDF of Score2, (e) Distribution of game results, (f) Histogram of Score1 and Score2 pooled togther.

5 Discussion

The nfl data set contains 13500 rows of observation with 8 columns for each row, and contains no missing values (NAs). Based on the numeric variables (score1, score2) located in Table 3, they have similar statistics, such as having the same interquartile range(IQR) and having similar standard deviations. It noticeable that score1 has a higher mean, median, and mode compared to score2. score1 has a mean 2.397 points higher than score2, a Median 2 points higher, and a Mode 3 points higher. The skewness values mean that there is a slight skewness and a right tail in the scores. Both score1 and score2 have a a kurtosis close to zero, score1 has a slightly higher peak and score2 is slightly flat.

Table 4 which focuses on the result shows that the home team ends up winning 56.65% of the time. This can also support why score1 has higher mean, median, and mode since score1 represents the home team. Table 6 represents how many home games Team1 has played and Table 7 represents how many away games Team2 has played. Table 8 shows the total count of games played for each team in the data set.

6 References

Data source: NFL dataset (nfl_data.csv). This dataset contains data collected and cleaned from NFL dataset from Kaggle on September 7, 2025.